

Agent Evaluation

Advanced AI Agents by Hand 🍌 Series - #3

Mohsena Ashraf

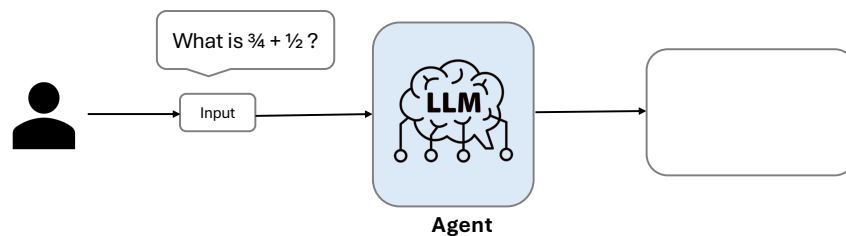
mentored by Prof. Tom Yeh



Hosted by:



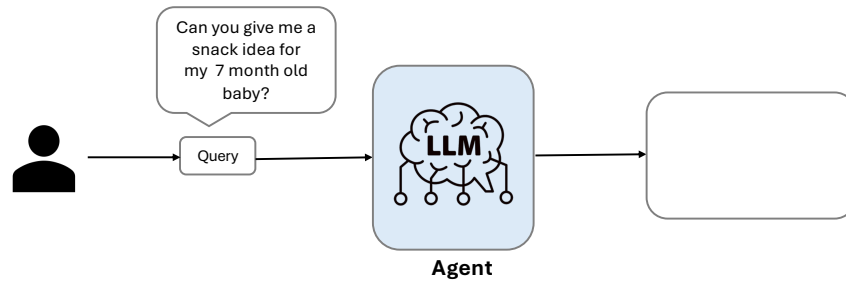
What is Agent Evaluation?



Hint: Fill in the blanks with adjectives/adverbs!

- Provides _____ answer?
- Falls _____?
- Avoids _____ actions?
- Handles _____ inputs?
- Cites _____ sources?

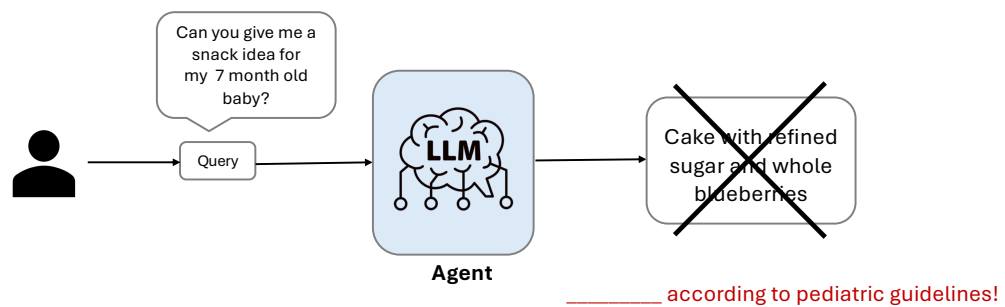
Why do it?



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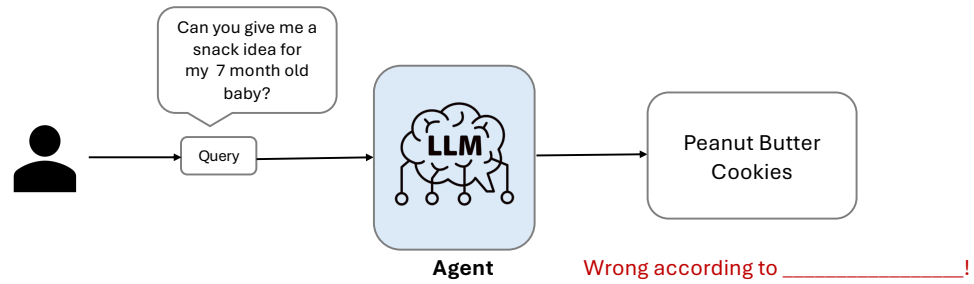
Why do it?



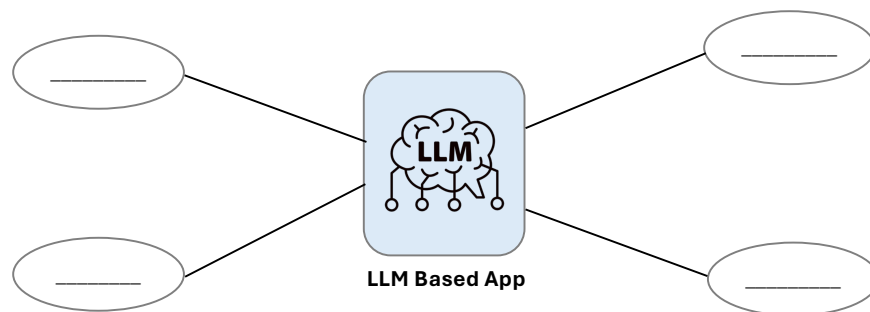
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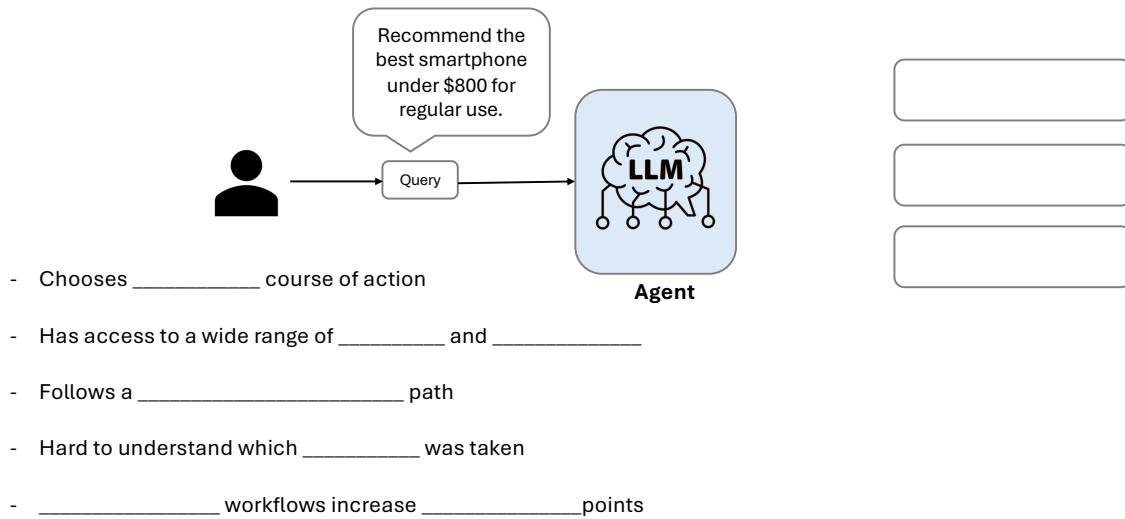
Another Mistake



Why it is Hard?



Why it is Hard?



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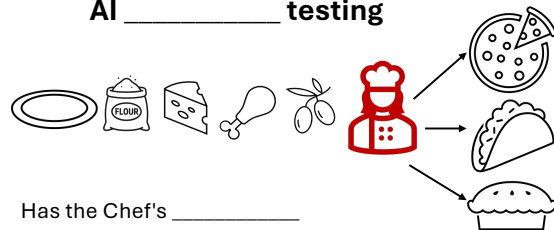
Difference?

Traditional _____ testing



- Following a recipe _____
- Fixed ingredients (_____), and steps (_____)
- Following correctly will always give the same dish (_____)
- Input + _____ -> Output
- _____ and predictable

AI _____ testing



- Has the Chef's _____
- _____ ingredients may lead to _____ dishes
- Chef adjusts based on intuition or experience (_____)
- Input + _____ -> _____ possible outputs
- Unpredictable

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Considerations in Agent Development

- Is the initial _____ correct?
- Do I have a proper _____?
- Do I need to change the _____?

Trial and Error?

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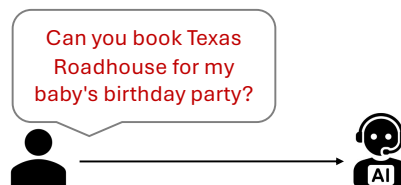
Evaluation Based Process?

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Agent Evaluation in Different Steps

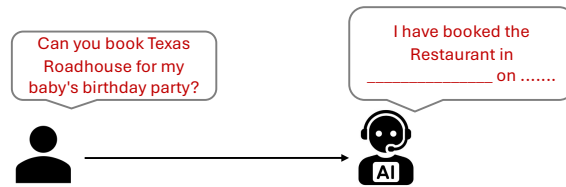


How do we evaluate step by step?

- _____: Did the agent choose the right _____ platform?
- _____: When it makes the booking, did it use the right _____?
- _____: Did it incorporate _____?
- _____: Does it present the _____ clearly and politely?
- _____: Is the output _____ factually correct?

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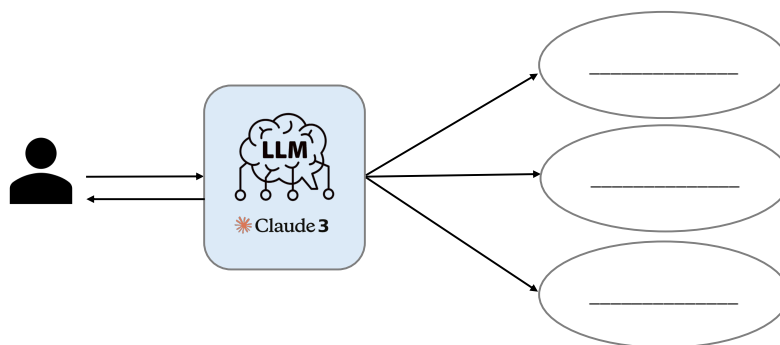
Why not only the Final Output?



- Agent might use _____ tool, _____ context, _____ tone
- Agent evals must be _____ after each _____
- Can use the _____ use cases as our _____ dataset

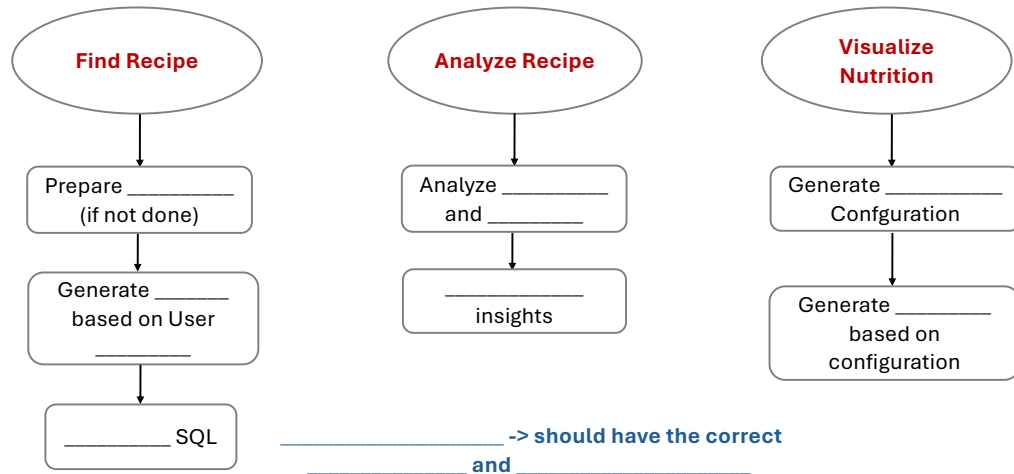
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Example Agent



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Functionalities of the Tools

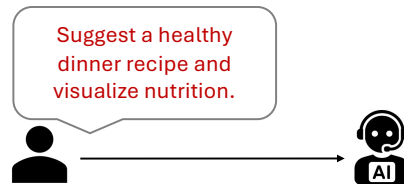


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Let's Observe our Agent!



- Complete _____ into every layer of our agent
- For monitoring the agent's _____ steps
- Like a _____
- prompt, response, token usage, api calls etc.



We could observe-

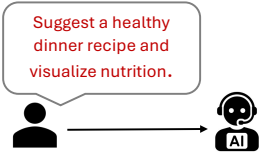
- **Find Recipe** -> _____ 5 candidate recipes
- **Analyze Recipe** -> Calculated _____ & _____
- **Visualize Nutrition** -> Generated a _____ chart of _____
- Makes it easier to _____ if the agent returns a _____ chart or a _____ output

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Observability is Made UP of -

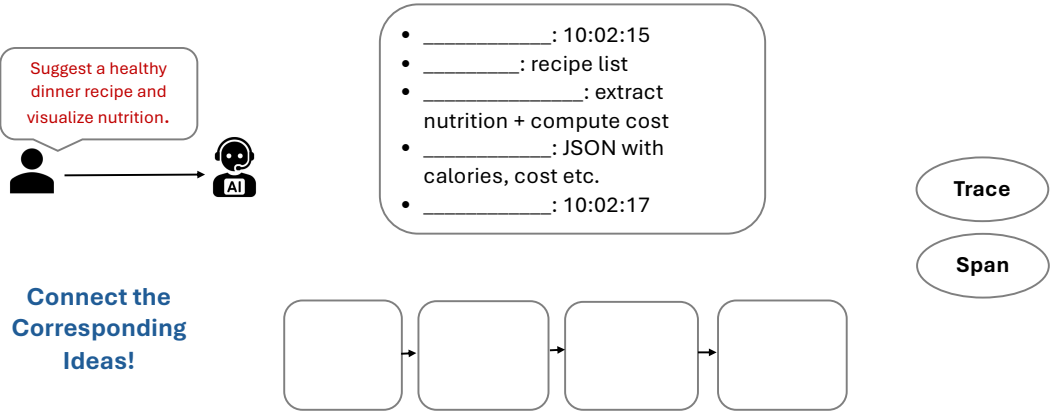
- Shows the entire chain of steps for one _____ request
- _____ run throughout the application
- From an _____ to all the way though to an _____

- A _____ unit of work inside a trace
- can be for _____ calls, for _____ calls, or just simple _____ flows
- _____ spans together form _____ trace



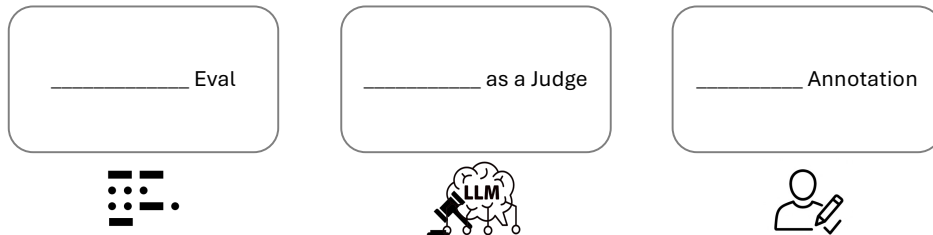
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Traces and Spans



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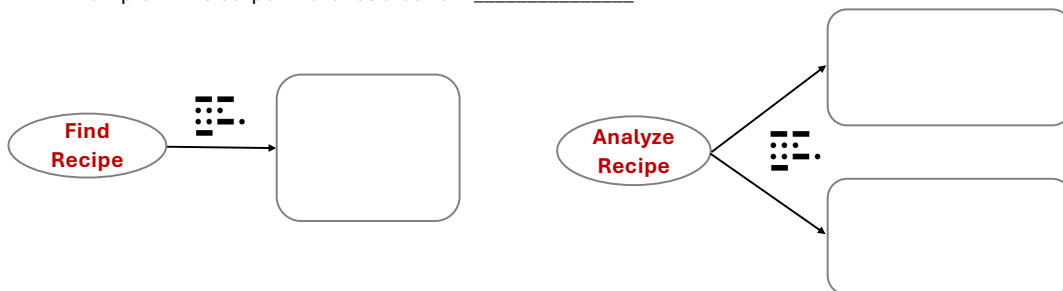
_____ Types of Evaluators



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Code Based Eval

- _____ way of evaluation
- Running codes on the _____ of the app
- Best suited when having _____ data, compares outputs to _____ outputs
- Comparison can be _____ or _____ match
- Example: If the output matches a certain _____



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Code Based Eval

```
def eval_find_recipe_output(_____, _____):  
    """  
    Code-based eval for the Recipe Lookup tool.  
    Ensures that no alcohol-based recipes are returned.  
  
    Arguments:  
        title (str): The title of the recipe  
        ingredients (list[str]): List of ingredients  
  
    Returns:  
        str: Pass/Fail with reason  
    """  
  
    banned_words = ["wine", "beer", "rum", "vodka", "whiskey", "brandy", "champagne"]  
  
    # Check recipe title  
    for word in banned_words:  
        if word.lower() in title.lower():  
            return f"Fail: Recipe title contains banned word '{word}'"  
  
    # Check ingredients  
    for ingredient in ingredients:  
        for word in banned_words:  
            if word.lower() in ingredient.lower():  
                return f"Fail: Ingredient contains banned word '{word}'"  
  
    return "Pass"
```

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LLM as a Judge

- Very _____ as able to run large scale evaluation in both _____ metrics
- A _____ LLM to judge the output
- A separate _____ for that LLM to evaluate based on some _____
- LLM assigns a _____

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Correct/Incorrect
Relevant/Irrelevant

Few things to keep in mind-

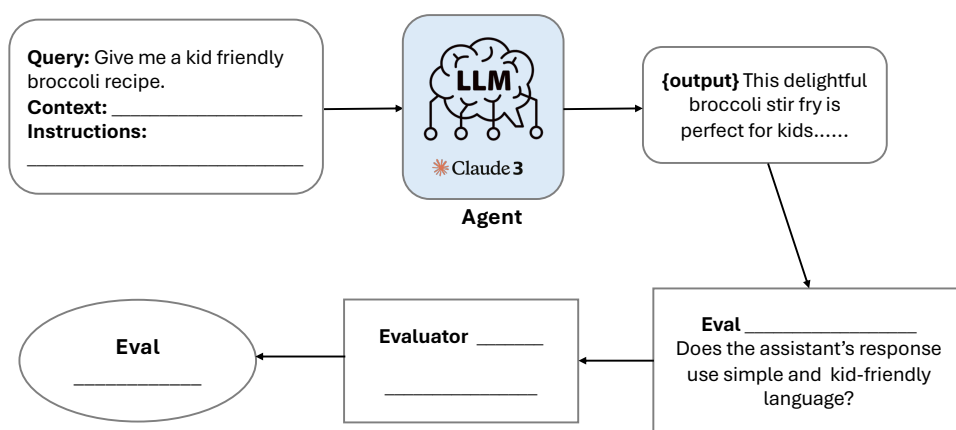
- Only _____ LLM models align closely to _____ models
- _____ 100% accurate
- adjusting the error margin by changing the _____ and the _____
- Use _____ classification labels instead of _____ scores

☐

score this response on a
scale from 1 to 10.

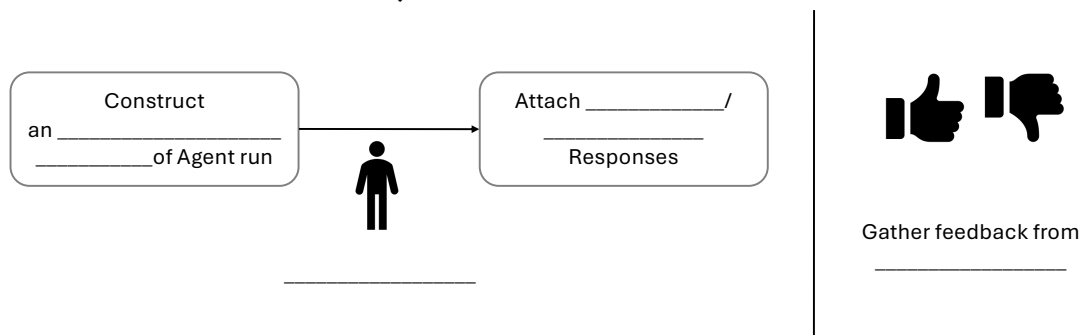
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LLM as a Judge



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Human Annotation

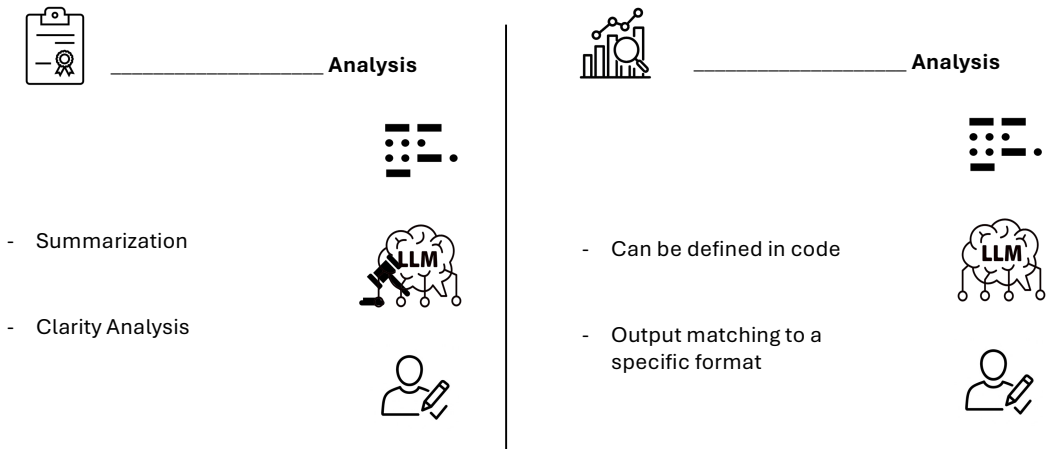


- _____ Evaluation Technique
- _____ at large scale as _____
- Selection _____ on choosing end users
- _____ to use at a large scale

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How to Select?

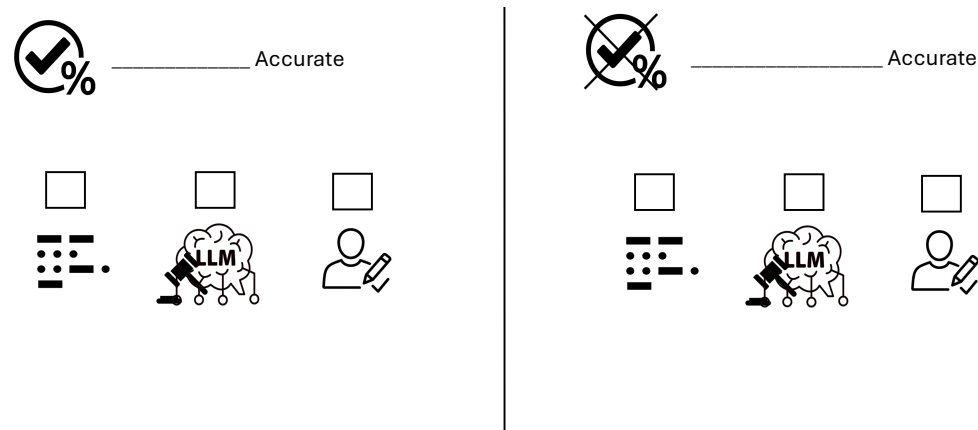
Hint: Connect the Corresponding Evals



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How to Select?

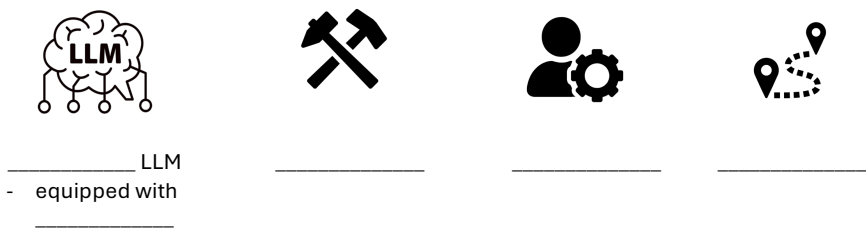
Hint: Tick the Corresponding Evals



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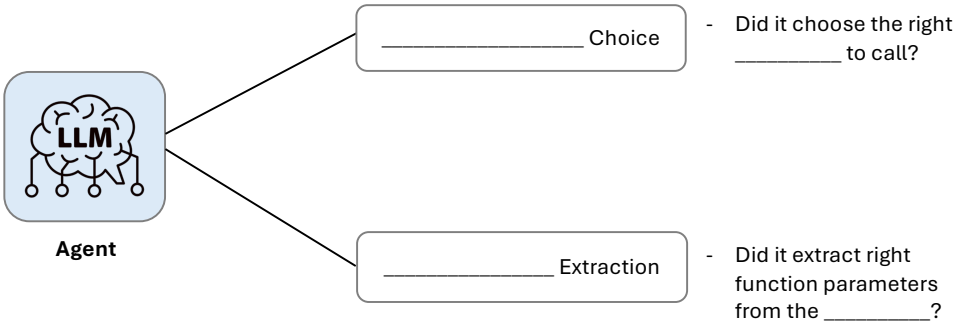
What things can we evaluate?

Different pieces of the _____ to apply these evaluation techniques ->



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Evaluating the Agent's LLM

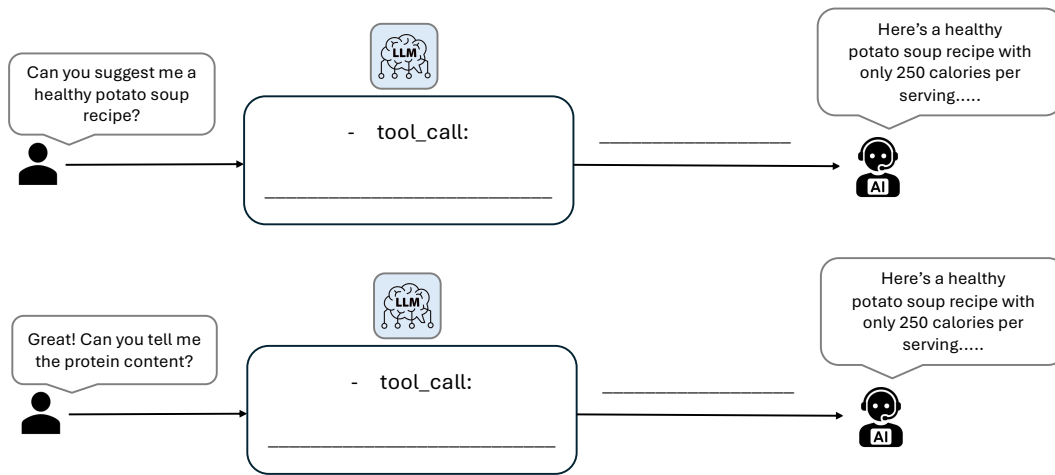


How to evaluate?

- We can use _____

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Why do we need Agent's LLM Evaluation?



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